

Computer Vision - Assignment # 02

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Problem 3; Rank of Given Matrices.

Matrix A: $\begin{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \\ \begin{bmatrix} 3 & 6 & 9 \end{bmatrix} \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 6 & 9 \end{bmatrix}$$

⇒ Relationship b/w rows:

$$\text{Row 2} = 3 \times \text{Row 1}$$

$$\begin{bmatrix} 3 & 6 & 9 \end{bmatrix} = 3 \times \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

⇒ Row Reduction.

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 6 & 9 \end{bmatrix} \Rightarrow \text{Original Matrix (Initial)}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow R_2 = R_2 - 3R_1$$

Hence, result / rank is one. $\text{Rank}(A) = 1$.

Matrix B: $\begin{bmatrix} [1, 2, 3] \\ [4, 5, 6] \\ [7, 8, 9] \end{bmatrix}$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \Rightarrow \text{Initial matrix}$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{bmatrix} \Rightarrow R_2 = R_2 - 4 \times R_1$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix} \Rightarrow R_3 = R_3 - 2 \times R_2$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -6 & -12 \end{bmatrix} \Rightarrow R_3 = R_3 + 6 \times R_2$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow R_3 = R_3 + 6 \times R_2$$

$\text{Rank}(B) = 2$

Problem 02; Prove that convolutions are commutative

$$a * b = b * a.$$

we need to show $(a * b)[n]$ is equal to $(b * a)[n]$.

$$(a * b)[n] = \sum_{k=-\infty}^{\infty} a[k] b[n-k].$$

definition: $(f * g)[n] = \sum_{k=-\infty}^{\infty} f[k] g[n-k].$

Suppose $m = n - k$, which means $k = n - m$.

When $k \rightarrow -\infty$, then $m \rightarrow +\infty$

When $k \rightarrow \infty$, then $m \rightarrow -\infty$ Also $dk = -dm$.

$$\begin{aligned} (f * g)[n] &= \sum_{k=-\infty}^{\infty} f[k] \times g[n-k] \\ &= \sum_{m=+\infty}^{-\infty} f[n-m] \times g[m] \times (-1) \\ &= - \sum_{m=+\infty}^{-\infty} f[n-m] \times g[m] \\ &= \sum_{m=-\infty}^{+\infty} f[n-m] \times g[m] \\ &= \sum_{m=-\infty}^{+\infty} g[m] \times f[n-m]. \end{aligned}$$

$$(g * f)[n] = \sum_{m=-\infty}^{+\infty} g[m] \times f[n-m].$$

$$(f * g)[n] = (g * f)[n] \text{ for all } n.$$

This proves, $f * g = g * f$

Hence,

$$a * b = b * a.$$

Problem 1; Calculate Gaussian Filters.

$$G(x) = \exp \left(-x^2 / 2\sigma^2 \right)$$

$$\frac{dG}{dx} = \frac{d}{dx} \left[e^{-x^2/2\sigma^2} \right]$$

$$\frac{dG}{dx} = e^{-x^2/2\sigma^2} \times \frac{d}{dx} \left(-x^2/2\sigma^2 \right)$$

$$dG/dx = e^{-x^2/2\sigma^2} \times -2x/2\sigma^2$$

$$\frac{dG}{dx} = -x/\sigma^2 \times e^{-x^2/2\sigma^2}$$

$$\frac{dG}{dx} = \frac{-x}{\sigma^2} \times G(x)$$

Second;

$$\frac{d^2G}{dx^2} = \frac{d}{dx} \left[\left(\frac{-x}{\sigma^2} \right) \times G(x) \right]$$

$$= - \left(\frac{1}{\sigma^2} \right) \times G(x) + \left(- \left(\frac{x}{\sigma^2} \right) \right) \times dG/dx$$

$$= - \left(1/\sigma^2 \right) \times G(x) + \left(x^2/\sigma^2 \right) \times G(x)$$

$$= \left[\left(x^2/\sigma^4 \right) - \left(1/\sigma^2 \right) \right] \times G(x)$$

$$= \frac{x^2 - \sigma^2}{\sigma^4} \times G(x)$$

$$G(x, y) = e^{-\frac{x^2 + y^2}{2\sigma^2}}$$

$$\frac{\partial G}{\partial x} = -\left(\frac{x}{\sigma^2}\right) \times G(x, y)$$

First derivative w.r.t. y.

$$\frac{\partial G}{\partial y} = -\frac{y}{\sigma^2} \times G(x, y)$$

Second,

$$x: \frac{\partial^2 G}{\partial x^2} = \frac{(x^2 - \sigma^2)}{\sigma^4} \times G(x, y)$$

$$y: \frac{\partial^2 G}{\partial y^2} = \frac{(y^2 - \sigma^2)}{\sigma^4} \times G(x, y)$$